

UrbanPulse — Mumbai Traffic Dataset Guide

A practical overview of how the 50-sensor Mumbai arterial traffic dataset was built

1. Why This Dataset Was Created

The original UrbanPulse model was trained on the METR-LA dataset from Los Angeles highways. To test how well the model transfers to a completely different city, we created a realistic synthetic dataset for **Mumbai, India**. Instead of trying to map thousands of narrow streets, we focused on Mumbai's 5 main arterial traffic corridors that carry over 80% of daily commuter traffic. This gives us a clean, realistic sandbox to test cross-city traffic predictions without expensive sensor hardware.

2. The 5 Corridors We Selected

We placed 50 virtual sensor checkpoints across Mumbai's main North-South highways and East-West connecting roads:

Corridor	Code	Stretch / Route	Length	Sensors	Normal Speed	Key Bottlenecks
Western Express Highway	WEH	Dahisar Toll Plaza to Bandra / Mahim	25.5 km	16 sensors	75-80 km/h	Kalanagar Jn, Jogeshwari
Eastern Express Highway	EEH	Thane Anand Nagar to Sion Circle	22.0 km	14 sensors	75-80 km/h	Chedda Nagar, Amar Mahal
Bandra-Worli Sea Link & Coastal	BWSL	Bandra Toll to Priyadarshini Park	10.5 km	6 sensors	85-90 km/h	Worli Sea Face merge
Jogeshwari-Vikhroli Link Road	JVLR	Connects WEH (Jogeshwari) to EEH (Vikhroli)	8.5 km	7 sensors	55-60 km/h	SEEPZ & Powai choke
Santacruz-Chembur Link Road	SCLR	Connects WEH (Milan) to EEH (Chedda Nagar)	6.5 km	7 sensors	50-55 km/h	Kurla Double Decker Flyover

3. How Road Connections Work (Graph Topology)

Traffic models need a graph to know which road connects to where. We connected sensors in two ways:

- **Along the same road:** Each sensor is linked to the next sensor down the highway (e.g. Dahisar -> Borivali -> Kandivali).
- **At highway junctions:** Where roads intersect in real life, we connected them in the graph. For example, JVLR connects to WEH at Jogeshwari and to EEH at Vikhroli. SCLR connects to WEH at Santacruz and EEH at Chembur. The Sea Link connects to WEH at Bandra.

Using real driving distances, we calculated connection weights so nearby sensors have strong influence on each other, creating a connected network of **524 directed road links**.

4. The 50 Sensor Checkpoints & Locations

Here are all 50 monitoring checkpoints with their real-world location names, GPS coordinates, highway distance marker, and normal open-road speed limit (in km/h):

Sensor ID	Corridor	Checkpoint Location Name	GPS Coordinates	Distance	Speed Limit
1001	WEH	Dahisar Toll Plaza	19.2550, 72.8625	0.0 km	80 km/h
1002	WEH	Borivali National Park	19.2310, 72.8600	2.7 km	80 km/h
1003	WEH	Kandivali Flyover	19.2080, 72.8580	5.3 km	80 km/h
1004	WEH	Malad Pushpa Park	19.1860, 72.8550	7.8 km	75 km/h
1005	WEH	Goregaon Hub Mall	19.1620, 72.8530	10.5 km	75 km/h
1006	WEH	Aarey Colony Exit	19.1480, 72.8520	12.1 km	75 km/h
1007	WEH	Jogeshwari JVLR Jn	19.1350, 72.8510	13.6 km	70 km/h
1008	WEH	Andheri Flyover	19.1190, 72.8490	15.4 km	70 km/h
1009	WEH	Vile Parle Airport	19.0980, 72.8480	17.8 km	75 km/h
1010	WEH	Santacruz Milan Flyover	19.0830, 72.8460	19.5 km	70 km/h
1011	WEH	Khar Subway Jn	19.0700, 72.8440	21.0 km	70 km/h
1012	WEH	Bandra Teachers Colony	19.0610, 72.8430	22.1 km	70 km/h
1013	WEH	Kalanagar Chokepoint	19.0550, 72.8460	22.9 km	65 km/h
1014	WEH	BKC Connector Entry	19.0570, 72.8550	23.9 km	65 km/h
1015	WEH	Bandra Terminus	19.0510, 72.8410	24.6 km	70 km/h
1016	WEH	Mahim Causeway North	19.0430, 72.8390	25.5 km	70 km/h
1017	EEH	Anand Nagar Toll Thane	19.1980, 72.9690	0.0 km	80 km/h
1018	EEH	Mulund Check Naka	19.1800, 72.9620	2.1 km	80 km/h
1019	EEH	Bhandup Sonapur	19.1570, 72.9490	4.9 km	80 km/h
1020	EEH	Kanjurmarg Flyover	19.1350, 72.9370	7.5 km	80 km/h
1021	EEH	Vikhroli JVLR Jn	19.1160, 72.9280	9.8 km	75 km/h
1022	EEH	Ghatkopar Pantnagar	19.0920, 72.9150	12.8 km	75 km/h
1023	EEH	Chedda Nagar Bottleneck	19.0740, 72.9020	15.1 km	65 km/h
1024	EEH	Amar Mahal Junction	19.0680, 72.8960	16.0 km	65 km/h
1025	EEH	Chembur Priyadarshini	19.0580, 72.8880	17.4 km	75 km/h
1026	EEH	Kurla Signal EEH	19.0510, 72.8810	18.5 km	70 km/h
1027	EEH	Everard Nagar	19.0450, 72.8750	19.4 km	75 km/h
1028	EEH	Chunabhatti Flyover	19.0400, 72.8680	20.3 km	75 km/h
1029	EEH	Sion Circle North	19.0360, 72.8620	21.2 km	70 km/h
1030	EEH	GTB Nagar Approach	19.0310, 72.8590	22.0 km	70 km/h
1031	BWSL	Bandra Toll Plaza	19.0410, 72.8220	0.0 km	85 km/h
1032	BWSL	Mid-Span Cable Stay	19.0270, 72.8160	2.2 km	90 km/h
1033	BWSL	Worli Sea Face South	19.0130, 72.8130	4.5 km	85 km/h
1034	BWSL	Worli Interchange	19.0060, 72.8110	5.4 km	85 km/h
1035	BWSL	Haji Ali Bridge	18.9830, 72.8090	8.0 km	85 km/h
1036	BWSL	Priyadarshini Park	18.9610, 72.8040	10.5 km	85 km/h
1037	JVLR	WEH Junction Start	19.1350, 72.8530	0.0 km	60 km/h
1038	JVLR	Majas Depot Jogeshwari	19.1330, 72.8670	1.5 km	60 km/h
1039	JVLR	SEEPZ Andheri East	19.1300, 72.8820	3.1 km	55 km/h
1040	JVLR	Powai Lake Promenade	19.1260, 72.8990	4.9 km	55 km/h
1041	JVLR	IIT Bombay Main Gate	19.1240, 72.9130	6.4 km	60 km/h
1042	JVLR	Gandhi Nagar Flyover	19.1200, 72.9230	7.6 km	60 km/h
1043	JVLR	EEH Junction End	19.1160, 72.9270	8.5 km	60 km/h
1044	SCLR	WEH Milan Start	19.0830, 72.8480	0.0 km	55 km/h
1045	SCLR	Vakola Junction	19.0790, 72.8580	1.1 km	50 km/h
1046	SCLR	BKC North Gate	19.0740, 72.8690	2.3 km	50 km/h
1047	SCLR	Kurla Double Decker	19.0680, 72.8790	3.6 km	50 km/h
1048	SCLR	Nehru Nagar Kurla	19.0640, 72.8890	4.8 km	55 km/h
1049	SCLR	Tilak Nagar ROB	19.0670, 72.8950	5.6 km	55 km/h
1050	SCLR	Chedda Nagar EEH End	19.0720, 72.9010	6.5 km	55 km/h

5. How the Traffic Speeds Were Calculated

Instead of just picking random numbers, speeds are generated based on real-world Mumbai driving patterns:

- **Morning Rush Hour (~9:30 AM):** Heavy traffic heading South toward business hubs (BKC and South Mumbai). Speeds drop heavily.
- **Evening Rush Hour (~7:00 PM):** Dispersal traffic heading North back to suburbs like Borivali and Thane.
- **Weekends:** No sharp morning rush; instead, smooth afternoon/evening traffic around shopping areas.
- **Notorious Bottlenecks:** Known choke points like Kalanagar Junction (Sensor 1013), Chedda Nagar (1023), and Kurla Flyover (1047) experience heavier slowdowns, dropping speeds down to 12–18 km/h during peak hours.
- **Monsoon Season (June to September):** Speeds drop by roughly 12% across the city to reflect heavy rain and waterlogging.
- **Natural Continuity:** Speeds carry momentum from minute to minute so the traffic curves look smooth and natural.

6. Weather Data Matching

Traffic is closely tied to weather. We created an hourly weather file matching Mumbai's coastal climate for the Santacruz area:

- **Temperature:** 22°C (winter mornings) to 36°C (summer afternoons).
- **Humidity:** Around 60% in winter, climbing up to 85–95% during the monsoon.
- **Visibility & Wind:** Normal 9–10 miles visibility, dropping during heavy rainstorms with typical coastal winds (10–30 km/h).

7. The 3 Output Files & How to Use Them

The generation script produces three exact files in `data/raw/` that match the project's data loader format:

File Name	What It Contains	Format / Size	Used By
MUMBAI-50.h5	Speed readings for all 50 sensors at 5-minute intervals (34,272 time steps)	HDF5 matrix (6 MB)	<code>data_loader.load_metr_la()</code>
adj_MUMBAI-50.pkl	Graph connections and distance weights between all 50 sensor nodes	Python Pickle (3-item tuple)	<code>data_loader.load_adj_mx()</code>
weather_MUMBAI_2019.csv	Hourly temperature, humidity, pressure, wind, and visibility records	CSV (3,000 rows)	<code>data_loader._parse_weather_csv()</code>

8. Running & Verifying

You can regenerate or verify the dataset at any time using two simple terminal commands:

- `python scripts/mumbai/generate_mumbai_dataset.py` — Synthesizes the 3 raw dataset files.
- `python scripts/mumbai/verify_mumbai_dataset.py` — Tests that the files load with zero errors.

9. References & Data Sources

- **TomTom Traffic Index (Mumbai Analysis):** Live and historical travel times, rush hour speeds (14–22 km/h), and citywide congestion metrics (53–61%):
<https://www.tomtom.com/traffic-index/city/mumbai/>
- **Mumbai Express Highway Corridors:** Route lengths (25.3 km WEH, 22 km EEH), lane configurations, and primary junction bottlenecks (Kalanagar, JVLR, Milan Subway, Chedda Nagar):
https://en.wikipedia.org/wiki/Western_Express_Highway | https://en.wikipedia.org/wiki/Eastern_Express_Highway
- **National Highway & Urban Speed Standards:** Speed limits and design velocities for multi-lane expressways (70–80 km/h) and urban connecting arterials (50–60 km/h):
https://en.wikipedia.org/wiki/Speed_limits_in_India
- **Open-Meteo Historical Weather API:** High-resolution ERA5 reanalysis archive for hourly temperature, relative humidity, pressure, and rain rates for Santacruz, Mumbai (19.076° N, 72.878° E):
<https://open-meteo.com/en/docs/historical-weather-api>
- **Climate of Mumbai (Santacruz Meteorological Profile):** Empirical climate data documenting coastal monsoon rainfall (June–Sept), 85–95% humidity peaks, and seasonal road flooding patterns:
https://en.wikipedia.org/wiki/Climate_of_Mumbai
- **METR-LA Benchmark & Graph Neural Network (Li et al., ICLR 2018):** Foundational spatial-temporal traffic forecasting framework, 5-minute sampling intervals, and Gaussian graph formulation:
<https://arxiv.org/abs/1711.05101> | <https://github.com/liyaguang/DCRNN>